Report on Assignment-2
Solving the Max-Cut Problem by GRASP

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May 12, 2025

1 Algorithm Descriptions

1.1 Randomized Algorithm

In the randomized approach, each vertex is randomly assigned to one of the two partitions with equal probability. This provides a baseline performance with minimal computational cost, but often results in suboptimal solutions due to lack of guidance.

1.2 Greedy Algorithm

The greedy algorithm begins by assigning the endpoints of the largest-weight edge to opposite partitions. Then, it iteratively places the remaining vertices to maximize the current cut weight. While efficient and intuitive, it may get stuck in local optima.

1.3 Semi-Greedy Algorithm

This is a hybrid of greedy and randomized techniques. Instead of always choosing the best vertex placement, it selects from a restricted candidate list based on a greedy function, introducing diversity to the solution and helping escape local optima.

1.4 Local Search

Starting from an initial solution (e.g., from the greedy or semi-greedy approach), this method explores nearby solutions by flipping vertex assignments to improve cut weight. It iterates until no further improvements are found, effectively fine-tuning the solution.

1.5 GRASP

GRASP (Greedy Randomized Adaptive Search Procedure) combines construction (semi-greedy) and local search phases in a multi-start iterative framework. Each iteration builds a solution and improves it via local search. The best result over all iterations is retained.

2 Comparison of Algorithms

Based on the experimental results, the following observations were made:

- The Randomized approach produced the lowest quality cuts due to its lack of structure, but required minimal computational effort.
- Greedy and Semi-Greedy methods performed significantly better than the Randomized algorithm. Semi-Greedy often outperformed Greedy by introducing controlled randomness into the selection process.
- Local Search substantially improved the quality of initial solutions through iterative refinements.
- GRASP consistently delivered the best results among all tested heuristics, effectively combining semi-greedy construction with local search.

3 Performance Overview

The following bar chart compares the cut values obtained by different algorithms on a sample of benchmark graphs:

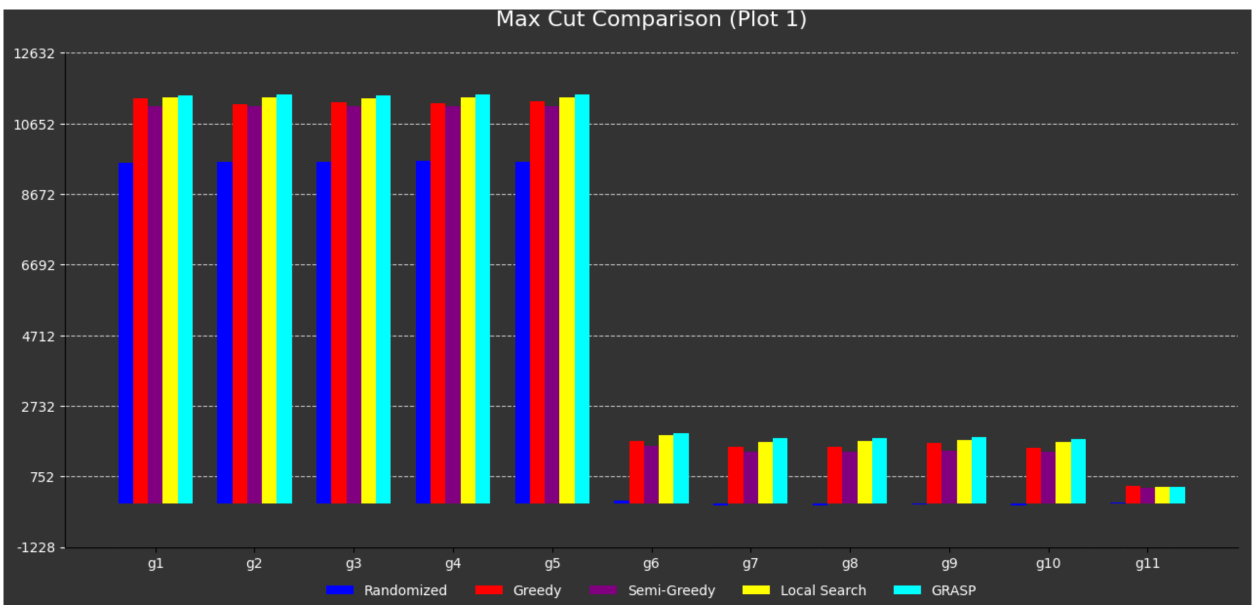
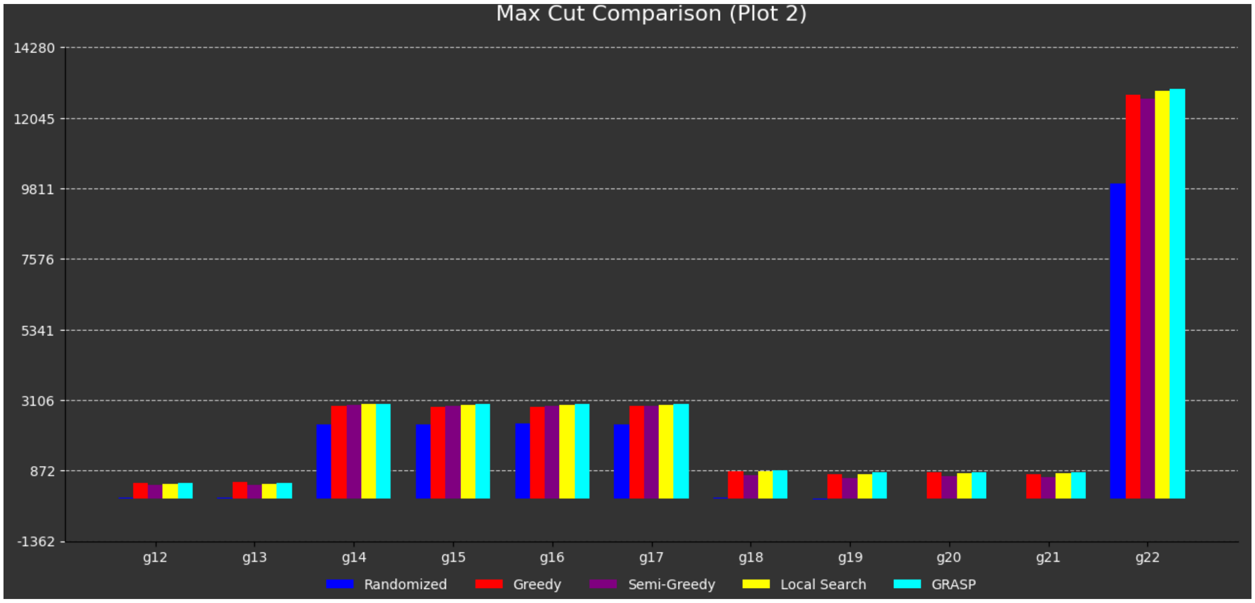
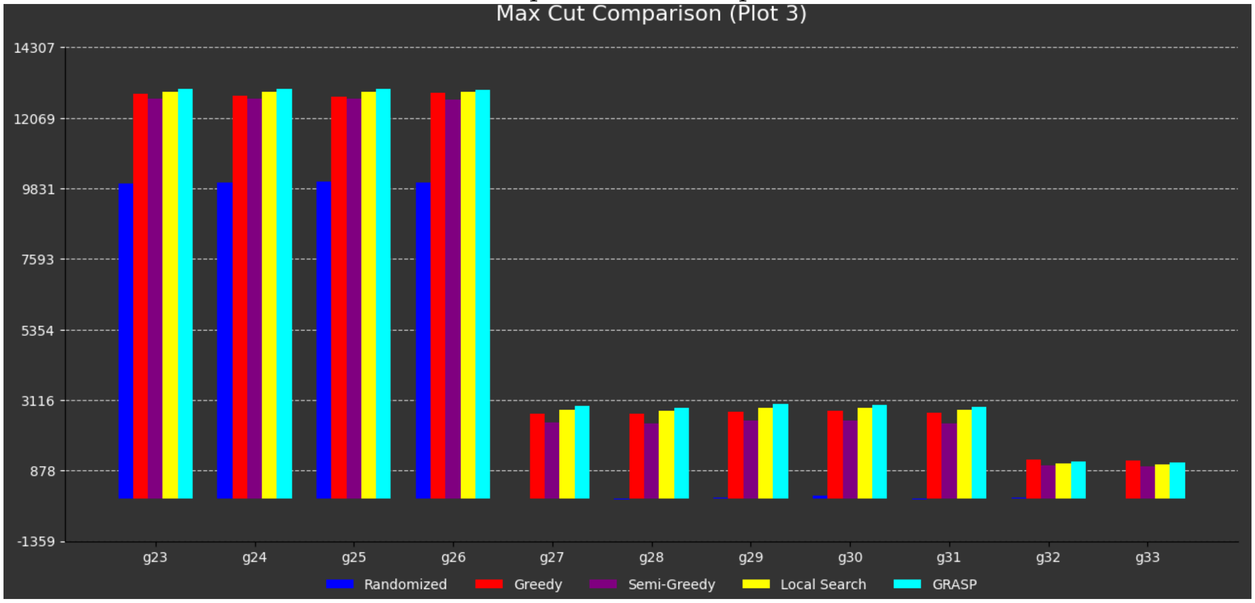


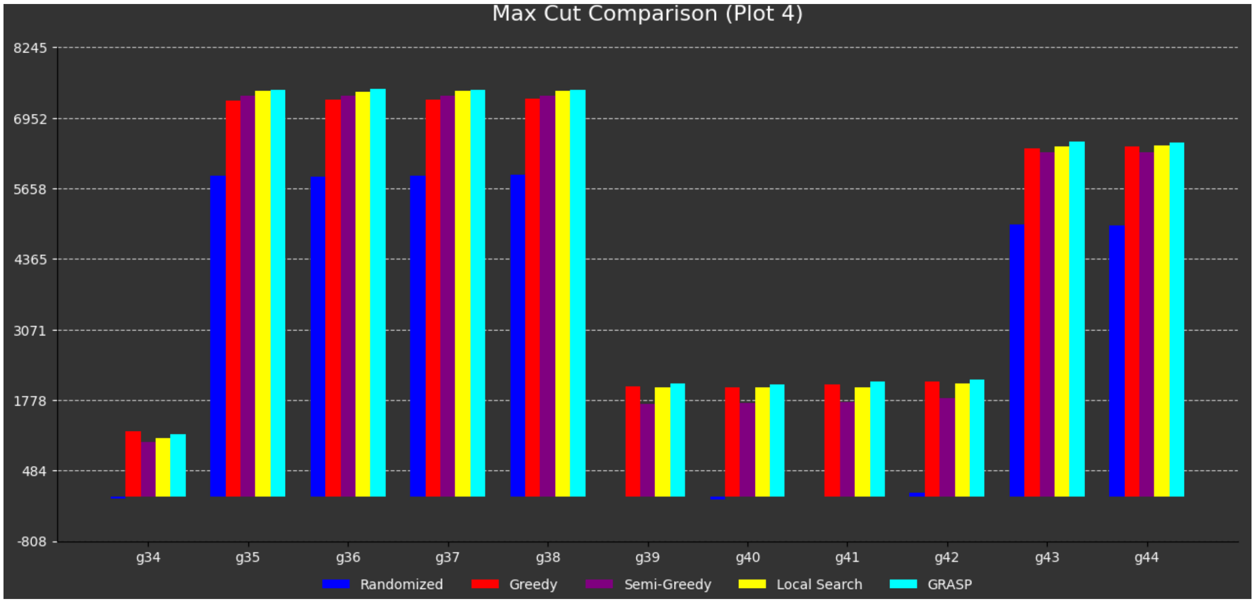
Figure 1: Max Cut Comparison for Graphs G1–G11



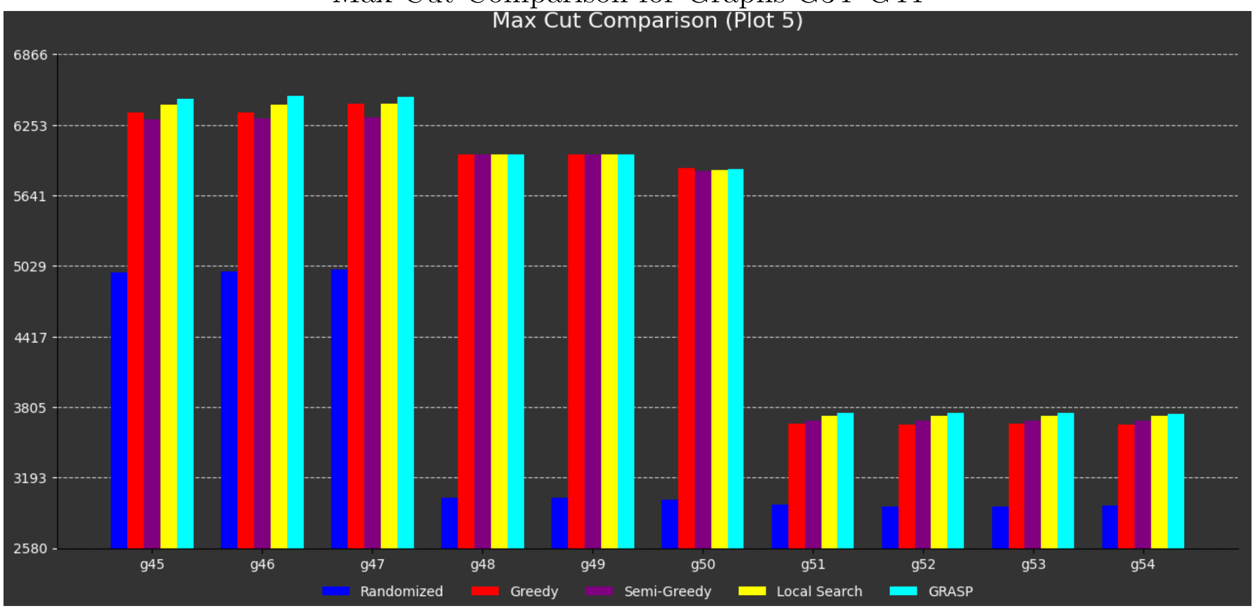
Max Cut Comparison for Graphs G12–G22



Max Cut Comparison for Graphs G23–G33



Max Cut Comparison for Graphs G34–G44



Max Cut Comparison for Graphs G45–G54

3.1 Observations

- The **Randomized** approach consistently underperforms compared to all others.
- **Greedy** and **Semi-Greedy** perform better than Randomized, but often plateau early.
- **Local Search** significantly improves upon initial greedy/semi-greedy solutions.
- **GRASP** consistently produces the highest cut values, outperforming all other algorithms across most graphs.
- However, none of the approaches reach the known best values for benchmark datasets.

4 Benchmark Results

Name	$\|V\|$	$\|E\|$	Randomized	Greedy	Semi-greedy	Local iters	Local avg	GRASP iters	GRASP best	Alpha	Known best
g1	800	19176	9560	11346	11141	6	11371	50	11445	0.500	12078
g2	800	19176	9586	11204	11143	7	11372	50	11463	0.500	12084
g3	800	19176	9582	11254	11150	7	11359	50	11440	0.500	12077
g4	800	19176	9592	11229	11149	7	11385	50	11477	0.500	0
g5	800	19176	9569	11277	11147	7	11372	50	11474	0.500	0
g6	800	19176	86	1742	1600	7	1907	50	1978	0.500	0
g7	800	19176	-73	1573	1433	7	1729	50	1827	0.500	0
g8	800	19176	-73	1586	1434	7	1733	50	1830	0.500	0
g9	800	19176	-23	1701	1479	7	1776	50	1843	0.500	0
g10	800	19176	-61	1556	1434	7	1726	50	1811	0.500	0
g11	800	1600	24	484	423	2	450	50	472	0.500	627
g12	800	1600	1	480	407	2	436	50	466	0.500	621
g13	800	1600	25	498	429	2	461	50	490	0.500	645
g14	800	4694	2339	2917	2939	2	2969	50	2990	0.500	3187
g15	800	4661	2329	2883	2918	2	2948	50	2968	0.500	3169
g16	800	4672	2350	2895	2921	2	2953	50	2979	0.500	3172
g17	800	4667	2337	2925	2921	2	2951	50	2981	0.500	0
g18	800	4694	7	846	729	3	837	50	884	0.500	0
g19	800	4661	-59	756	649	3	761	50	825	0.500	0
g20	800	4672	-24	813	686	3	780	50	823	0.500	0
g21	800	4667	-30	764	665	4	775	50	821	0.500	0
g22	2000	19990	9982	12773	12656	5	12890	50	12977	0.500	14123
g23	2000	19990	9999	12827	12675	5	12913	50	13002	0.500	14129
g24	2000	19990	10019	12780	12675	5	12907	50	13001	0.500	14131
g25	2000	19990	10035	12752	12666	5	12903	50	12989	0.500	0
g26	2000	19990	10007	12852	12663	5	12894	50	12970	0.500	0
g27	2000	19990	-18	2694	2404	7	2813	50	2932	0.500	0
g28	2000	19990	-54	2682	2376	6	2780	50	2874	0.500	0
g29	2000	19990	9	2739	2454	6	2878	50	2981	0.500	0
g30	2000	19990	79	2778	2471	7	2874	50	2955	0.500	0
g31	2000	19990	-31	2699	2372	6	2798	50	2904	0.500	0
g32	2000	4000	8	1226	1044	3	1112	50	1154	0.500	1560
g33	2000	4000	-25	1210	1013	2	1088	50	1134	0.500	1537
g34	2000	4000	-20	1210	1010	2	1087	50	1146	0.500	1541
g35	2000	11778	5891	7264	7366	2	7442	50	7474	0.500	8000
g36	2000	11766	5873	7294	7356	3	7433	50	7491	0.500	7996
g37	2000	11785	5891	7296	7366	3	7442	50	7474	0.500	8009
g38	2000	11779	5902	7298	7362	3	7441	50	7463	0.500	0
g39	2000	11778	0	2026	1707	4	2009	50	2074	0.500	0
g40	2000	11766	-54	2009	1729	4	2008	50	2068	0.500	0
g41	2000	11785	7	2058	1735	4	2010	50	2124	0.500	0
g42	2000	11779	83	2115	1807	4	2086	50	2158	0.500	0
g43	1000	9990	4997	6390	6315	5	6437	50	6524	0.500	7027
g44	1000	9990	4986	6428	6315	5	6441	50	6499	0.500	7022
g45	1000	9990	4975	6366	6302	5	6430	50	6480	0.500	7020
g46	1000	9990	4984	6362	6310	5	6433	50	6509	0.500	0
g47	1000	9990	5000	6437	6320	5	6437	50	6496	0.500	0
g48	3000	6000	3016	6000	6000	1	6000	50	6000	0.500	6000
g49	3000	6000	3019	6000	6000	1	6000	50	6000	0.500	6000
g50	3000	6000	3005	5880	5858	1	5859	50	5874	0.500	5988
g51	1000	5909	2962	3661	3690	2	3731	50	3753	0.500	0
g52	1000	5916	2945	3653	3690	2	3730	50	3751	0.500	0
g53	1000	5914	2938	3660	3686	2	3727	50	3757	0.500	0
g54	1000	5916	2948	3656	3689	2	3727	50	3748	0.500	0

5 Conclusion

GRASP consistently outperformed all other algorithms by effectively combining construction and local search phases. While simpler methods like Randomized and Greedy provide quick approximations, they lack the optimization depth that GRASP achieves. The performance gap between GRASP and known optimal solutions suggests room for further improvements through parameter tuning.